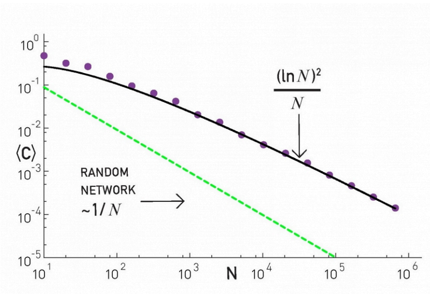
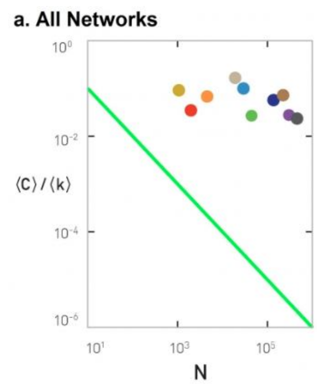
L5: Clustering in PA Model

Lesson-4 described the Preferential Attachment *(PA)* model and we saw that it generates networks with a power-law degree distribution *(with exponent=3)*. Are those networks also ***small-world,*** the way we have defined this property in this Lesson?



The plot at the left shows the average clustering coefficient for networks of increasing size *(the number of nodes is shown here as N).* The networks are generated with the PA model, with *m=2* links for every new node. Note that the clustering coefficient is significantly higher than that of random *G(N,p)* random graphs of the same size and density (*p=m/(N-1)*). It turns out (even though we will not prove it here) that the average clustering coefficient of a PA network scales as $\frac{(\ln N)^2}{N}$ – this is much larger than the corresponding clustering coefficient in *G(N,p)*, which is $O(\frac{1}{N})$.

However, this also means that the clustering coefficient of PA networks decreases with the network size *N*. This is not what we have seen in practice *(see figure at the right)*. In most real-world networks, the clustering coefficient does not reduce significantly at least as the network grows. Thus, even though the PA model produces significant clustering relative to random networks, it does not produce the clustering structure we see in real-world networks.



Food For Thought

Can you derive the mathematical expression we give here for the clustering coefficient of PA networks?